Jaehyun Ha

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Research Interests

Building AI-native database systems for unified analytics and inference on multi-modal data

- **Graph Databases:** Architecting a foundational query framework based on a unified graph representation of multi-modal data
- **Semantic Operator:** Embedding Large Language Models (LLMs) into the database to enable context-aware querying of unstructured data
- **Query Optimization:** Designing proxy models and novel cardinality estimation methods to optimize the execution of computationally expensive semantic operators

Education

POSTECH, Graduate School of Artificial Intelligence — Mar. 2022 – Present
M.S./Ph.D. Integrated Program — (Expected: Feb. 2028)
Data Systems Lab
Advisor: Prof. Wook-Shin Han

POSTECH, Dept. of Computer Science and Engineering — Mar. 2018 – Mar. 2022
B.S. in Computer Science and Engineering
GPA: 4.06 / 4.3
*Graduated with the **highest** GPA in CS Dept.*

Ulsan Science High School — Mar. 2016 – Mar. 2018

Professional Experiences

Google Systems Research Group, Sunnyvale, CA, USA — Jun 2026 – Sep 2026
Visiting Student Researcher

- Developed TPU-native relational operators and explored hardware–software co-design for analytical and semantic query processing
- Hosted by Yeounoh Chung; mainly collaborated with Helena Caminal and Fatma Özcan

University of Illinois Urbana-Champaign (UIUC), IL, USA — Jun 2025 – Sep 2025
Visiting Scholar

- Investigated intent-driven planning and optimization of semantic operators for efficient LLM-powered query processing
- Worked with Prof. Yongjoo Park

SK Hynix Digital Platform Team, Icheon, Korea — Jul 2020 – Aug 2020
Software Engineer Intern

- Developed real-time video comment overlay system for SK hynix's internal video streaming systems
- Won silver prize (2nd) in internship contest

Publications

1. **Jaehyun Ha**, Yeounoh Chung, Helena Caminal, Aditya K Kamath, Fatma Özcan, Arvind Krishnamurthy, Henry M. Levy. *The Case for TPU-Accelerated Analytical Query Processing.* Submitted to VLDB 2027.

2. **Jaehyun Ha**, Yongjoo Park, Wook-Shin Han. *CADENZA: Compiling Natural-Language Intent into Task-Specific Operator DAGs for Semantic Query Processing.* SIGMOD 2027.

3. **Jaehyun Ha**, Yongjoo Park, Wook-Shin Han. *CADENZA in Action: Breaking the Monolith with Intent-Dependent Plan Spaces for Semantic Queries.* VLDB 2026 (Demo).

4. Taesung Lee, **Jaehyun Ha**, Byungchul Tak, Wook-Shin Han. *TurboLynx: Schemaless Graph Engine Strikes Back for General-Purpose Analytics.* VLDB 2026.

5. Taesung Lee, **Jaehyun Ha**, Byungchul Tak, Wook-Shin Han. *TurboLynx in Action: A Schemaless Graph Engine for General-Purpose Analytics.* VLDB 2026 (Demo).

6. Wonseok Lee[*], **Jaehyun Ha**[*], Wook-Shin Han, Changgyoo Park, Myunggon Park, Juhyeng Han, Juchang Lee. ([*]equal contribution) *DoppelGanger++: Towards Fast Dependency Graph Generation for Database Replay.* SIGMOD 2024.

7. Wonseok Lee, **Jaehyun Ha**, Wook-Shin Han, Changgyoo Park, Myunggon Park, Juhyeng Han. *DoppelGanger++ in Action: A Database Replay System with Fast Dependency Graph Generation.* VLDB 2024 (Demo).

8. Kyoungmin Kim, **Jaehyun Ha**, George Fletcher, Wook-Shin Han. *Guaranteeing the O(AGM/OUT) runtime for uniform sampling and size estimation over joins.* PODS 2023.

Invited Talks

- Optimizing Semantic Queries: Current Work and Open Questions — Cornell University — Sep 2026
- CADENZA: Intent-Dependent Planning for Semantic Queries — Google Systems Research Group — Jan 2026

Projects

TPU-Accelerated Analytical Query Processing — Jun 2026 – Sep 2026

Visiting Student Research at Google Systems Research Group

- **Problem:** Semantic queries combine relational processing with LLM inference, but execution across CPUs and AI accelerators incurs costly data movement. Running both on TPUs requires adapting data-dependent relational operations to a static execution model
- **Contribution:** NTA introduces TPU-native relational operators and hardware-aware physical planning in JAX/XLA, enabling the first execution of all 22 TPC-H queries entirely on TPUs and laying the groundwork for unified relational and semantic query processing
- **Impact:** Achieved geometric-mean speedups over DuckDB of 7.0× at 100GB on one TPU and 68.6× at 1TB on 16 TPUs, excluding compilation and data transfer
- **Outcome:** Submitted to VLDB 2027

Reliability Evaluation of Semantic Query Processing — Feb 2026 – May 2026

- **Problem:** LLM-backed query processing requires reliable outputs across diverse analytical tasks, yet evaluating correctness and robustness remains challenging
- **Contribution:** Developed an evaluation methodology to assess the reliability of semantic query processing and analyze failure modes across query workloads
- **Impact:** Identified reliability limitations and informed validation strategies for more dependable semantic query processing

Intent-Driven Semantic Operator Optimization Jun 2025 – Jan 2026

Visiting Research with Prof. Yongjoo Park, UIUC

- **Problem:** Treating semantic operators as monolithic model calls limits optimization, leaving substantial trade-offs between output quality, inference latency, and monetary cost
- **Contribution:** CADENZA compiles natural-language intent into task-specific operator DAGs, exposing intermediate results to relational optimization and jointly tuning physical implementations
- **Impact:** Reduced scenario-average latency and monetary cost by up to 165.7× and 310.3×, respectively, against state-of-the-art systems on SemBench
- **Outcome:** Accepted at SIGMOD 2027 and VLDB 2026 (Demo)

High-Performance Schemaless Graph Database System Jan 2023 – May 2025

- **Problem:** Schema heterogeneity in property graphs creates storage, execution, and optimization overheads that limit analytical query performance
- **Contribution:** TurboLynx integrates graphlet-based columnar storage, schema-aware query execution, and cost-based optimization to efficiently support both graph traversal and analytical workloads
- **Role:** Core system developer; authored approximately 50% of commits in a 246K+ LOC codebase. Coded primarily without LLM assistance in 2023–2024
- **Impact:** Outperformed graph DBMSs by up to 183.9× and relational DBMSs by up to 41.27× across LDBC SNB Interactive, TPC-H, and DBpedia evaluations
- **Outcome:** Accepted at VLDB 2026 and VLDB 2026 (Demo)
- **Website:** https://turbolynx.io/

High-Speed Dependency Graph Generation for Database Replay Dec 2022 – Dec 2023

Joint Research with SAP Labs Korea

- **Problem:** Dependency graph generation bottlenecks database replay, limiting efficient testing of system changes on captured production workloads
- **Contribution:** DoppelGanger++ introduces a stateful single forward scan algorithm and its parallel extension to efficiently construct dependency graphs that preserve replay correctness
- **Impact:** Implemented in SAP HANA, bringing the research into a commercial database system
- **Outcome:** Accepted at SIGMOD 2024 and VLDB 2024 (Demo)

Provably Efficient Join Sampling and Cardinality Estimation Mar 2022 – Nov 2022

- **Problem:** Generating training samples from relational data can require prohibitively large joins; existing sampling methods incurred additional input-size overheads for general joins
- **Contribution:** Degree-based rejection sampling (DRS) achieves $\widetilde{O}(\mathrm{AGM}/\mathrm{OUT})$ time for uniform sampling and cardinality estimation over arbitrary joins after linear-time index construction
- **Impact:** Resolves an open problem in join sampling and estimation by eliminating the extra input-size factor, supporting training-data generation without materializing full join results
- **Outcome:** Accepted at PODS 2023; the first PODS paper with authors affiliated with a Korean institution

Technical Skills

Programming Languages	C/C++, Python, SQL, CUDA
Database Systems	PostgreSQL, DuckDB, Kuzu, Neo4j
Systems & Build Tools	Linux, Docker, CMake, Bazel
ML Frameworks	PyTorch, Hugging Face

Honors and Fellowships

- POSTECHIAN Fellowship 2022, 2023, 2024, 2025
- SK hynix Industry-Sponsored Scholar 2020–Present
- Samsung Dream Scholarship Foundation Scholar 2019-2021

Teaching Experiences

- **Teaching Assistant**, Agentic RAG, Samsung Electronics — Spring 2026
- **Teaching Assistant**, AIGS540: Big Data Processing, POSTECH — Spring 2024
- **Teaching Assistant**, CSED421: Database Systems, POSTECH — Spring 2022

References

Wook-Shin Han wshan@dblab.postech.ac.kr
POSTECH Distinguished Professor, Pohang University of Science and Technology (POSTECH), Korea

Yongjoo Park yongjoo@g.illinois.edu
Assistant Professor, Department of Computer Science, University of Illinois Urbana-Champaign (UIUC), USA